

Technical Report: Analysis of the Downloaded Catalysis-Hub CO2 Dataset

June 6, 2026

Audience: experimental chemists interested in the interpreted trends rather than the theoretical implementation details. **Source file analyzed:** /Users/dk2994/Desktop/git/PFUI/notebooks/catalysis_hu

1 Executive Summary

This report analyzes the local Catalysis-Hub HDF subset that was previously downloaded and converted into a tabular format. The dataset contains **133 linked system rows** corresponding to **40 unique reaction entries** across **16 surface compositions** from **6 literature sources**. The chemical focus of the downloaded subset is **CO2 activation on surfaces**. The dominant reaction classes are:

- molecular CO2 adsorption, $\text{CO}_2(\text{g}) + * \rightarrow \text{CO}_2^*$
- dissociation toward $\text{CO}^* + \text{O}^*$
- deep deoxygenation toward C^*
- first hydrogenation toward HCOO^* , COOH^* , and CH_2O^*

The main data-driven conclusions are:

1. **Vacancy-rich ceria-derived oxide surfaces bind CO2 most strongly.**

The most stabilizing adsorption energies in the dataset are found for **CeO2-rocksalt-10vacancy**, **SmCeO2-rocksalt-10vacancy**, and **CeO2-rocksalt-20vacancies**, with reconstructed CO2 adsorption energies between roughly -2.84 eV and -1.12 eV.

1. **Ni binds CO2 much more weakly than reduced ceria, and facet matters.**

In this subset the Ni entries shift from slightly exergonic adsorption on Ni(211) (-0.161 eV) to mildly endergonic adsorption on Ni(111) (+0.325 eV).

1. **Direct C formation from CO2 is strongly uphill on all surfaces where it appears.**

The $\text{CO}_2 \rightarrow \text{C}^*$ entries are among the most endergonic reactions in the entire dataset, typically between +3.95 eV and +5.78 eV. This strongly argues against deep direct deoxygenation as a realistic low-energy route under the conditions represented here.

1. ****For the small metal subset with hydrogenation data, HCOO^* is consistently more stable than COOH^* .****

On Ag, Au, Pd, and Rh the energy difference $E(\text{HCOO}^*) - E(\text{COOH}^*)$ is always negative, meaning the formate-like branch is favored over the carboxyl branch in this dataset.

1. **Barrier information is sparse but chemically informative.**

Only six unique reactions in the subset contain a finite activation barrier. Among them, Ru and Co are the only surfaces where the direct $\text{CO}_2 \rightarrow \text{CO}^* + \text{O}^*$ dissociation step is both thermodynamically favorable and associated with a comparatively small barrier.

2 1. What is in the downloaded file?

The local HDF is not a single homogeneous study. It is a stitched subset of Catalysis-Hub entries from several publications, different DFT engines, and different methodological settings. That is useful for trend spotting, but it means the dataset should be treated as a **comparative screening dataset**, not as a single internally uniform benchmark.

metric	value
system rows in HDF	133
unique reactions	40
surface compositions	16
publications represented	6
rows marked review by curation	8
directly reconstructable CO_2 adsorption energies	9
finite activation barriers	6

2.1 Publications and methods represented

The downloaded subset spans classic metal-surface CO_2 dissociation studies, oxide-vacancy chemistry, and small electrochemical CO_2 -reduction descriptor datasets. The method mix is dominated by VASP / PBE+U for the ceria-based oxide entries, DACAPO / RPBE for several older metal dissociation entries, and QE / BEEF-vdW for the hydrogenation branch entries on close-packed metals.

3 2. Curation and data quality

A light curation pass was applied before interpretation. Rows were marked for review when they lacked an energy, had no reliable structure, or exceeded conservative force thresholds. In practice, the review rows are almost entirely the six N/A entries that only carry a reaction summary without a linked system energy, plus two rows with somewhat elevated residual forces. This matters because the reaction-level trends remain usable, but **system-level adsorption-energy reconstruction is only meaningful when the corresponding gas, surface, and adsorbate rows are all present under the same reaction id**.

4 3. Reaction classes represented in the subset

The figure below shows that the file is heavily centered on CO_2 adsorption and first-step transformations rather than on long, fully resolved pathways. This is exactly what an experimental reader should keep in mind: this dataset is strong for **first mechanistic ranking questions** such as “which surfaces bind CO_2 strongly?”, “which direction of the first hydrogenation step is preferred?”, and “is direct dissociation plausible?”. It is not a complete mechanistic map all the way to a final product.

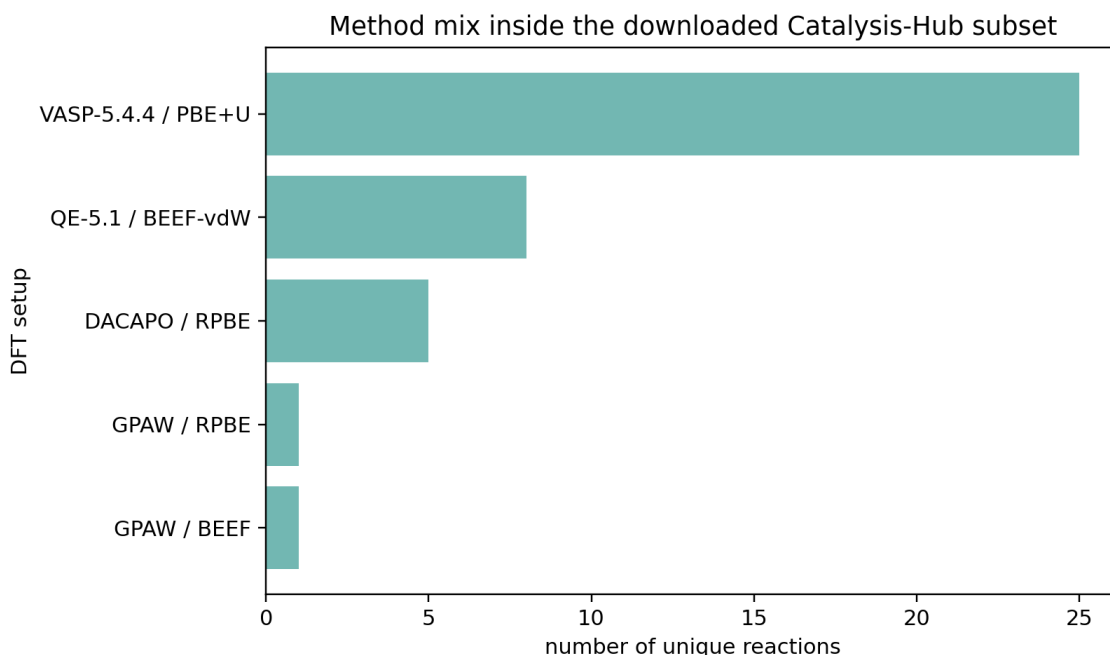


Figure 1: *
Method mix

5 4. Reaction-energy landscape across surfaces

The heatmap below condenses the core result of the dataset. Blue entries are stabilizing or exergonic, red entries are uphill. Three broad patterns emerge:

- **Oxide and defective oxide surfaces** stabilize molecular CO₂ adsorption much more strongly than the small metal subset.
- **Direct CO formation** from CO₂ is uphill on the oxide entries shown here, but can become favorable on specific stepped transition-metal surfaces such as Ru(211) and Co(211).
- **Direct carbon formation** is always strongly uphill and therefore chemically implausible as a facile elementary step in this dataset.

5.1 Surface-family summary

surface_family	reactions	mean_reaction_energy	min_reaction_energy	max_reaction_energy	barriers
metal	19	0.824	-1.477	4.947	5
oxide/support	20	1.632	-2.843	5.778	1
single-atom/support	1	0.412	0.412	0.412	0

The table above should not be over-read quantitatively because the families are sampled unevenly, but qualitatively it is clear that:

- the **oxide/support** class contains the strongest CO₂-binding entries;
- the **metal** class contains the most direct dissociation/barrier data;
- the single Fe–N–C entry is too sparse for a broad comparative statement.

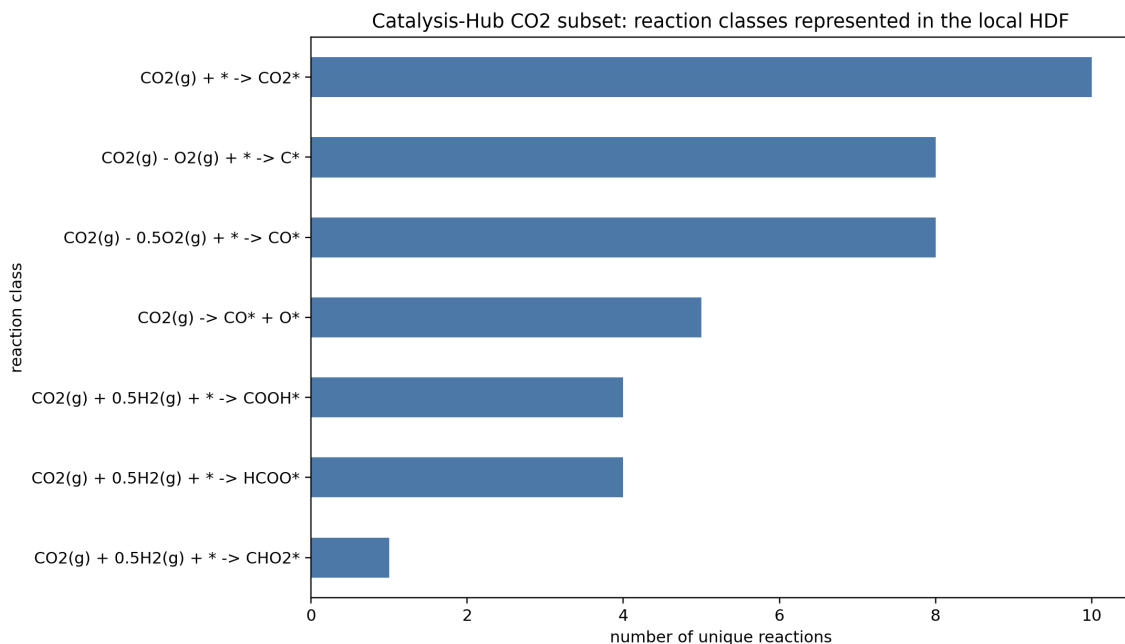


Figure 2: *
Equation counts

6 5. Directly reconstructable CO2 adsorption energies

A valuable technical check is that the local HDF contains enough information to reconstruct adsorption energies directly for the `CO2*` entries. For nine reactions, the corresponding `CO2gas`, `star`, and `CO2star` rows are present under the same reaction id. Using $E_{\text{ads}}(\text{CO}_2) = E(\text{CO}_2^*) - E(*) - E(\text{CO}_2\text{g})$ reproduces the published Catalysis-Hub `reactionEnergy` values to numerical precision.

6.1 Experimental interpretation

For an experimental audience, the practical reading is:

- **strongly negative adsorption energies** imply surfaces that are effective at trapping and activating CO_2 ;
- **mildly negative or near-zero adsorption energies** imply weaker molecular binding, often more compatible with facile desorption or with the need for additional activation steps;
- **positive adsorption energies** mean that molecular CO_2 adsorption itself is not favored under the chosen reference.

In this subset, vacancy-rich ceria surfaces clearly dominate the strong-binding regime, whereas Ni lies much closer to thermoneutral adsorption.

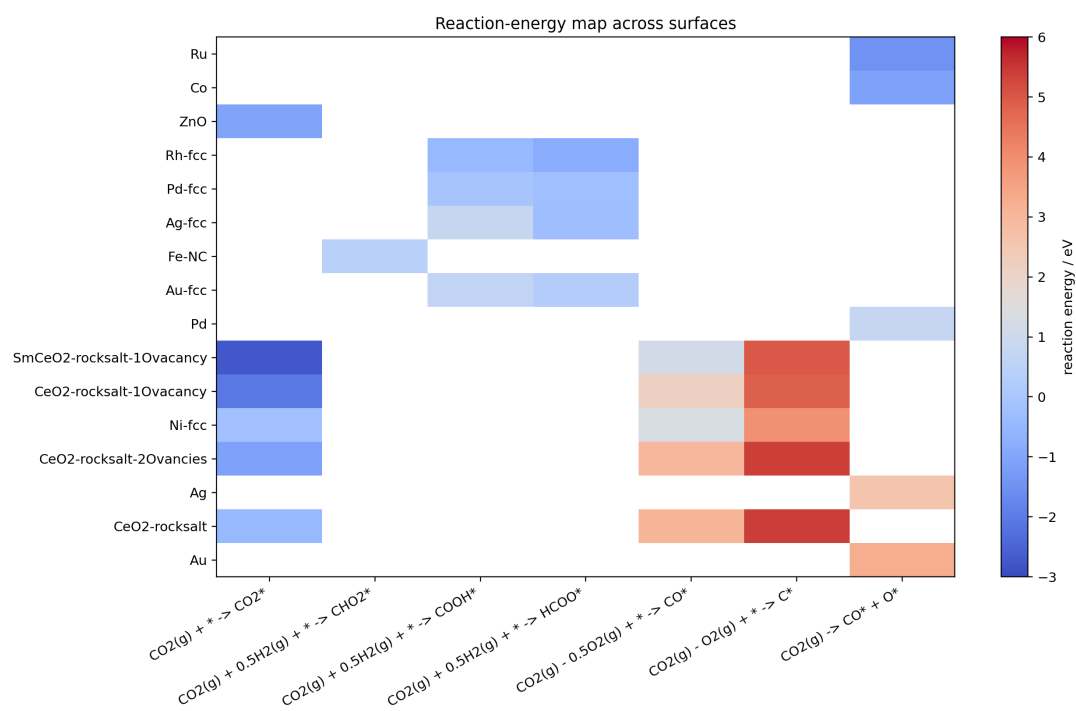


Figure 3: *
Reaction heatmap

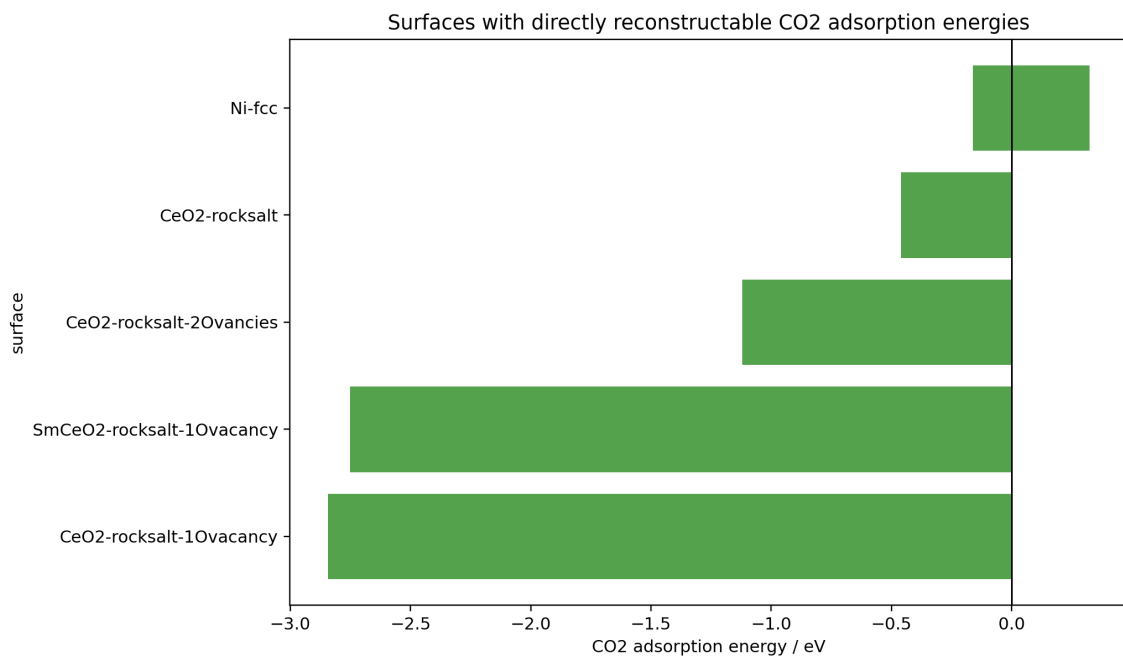


Figure 4: *
CO2 adsorption

6.2 Lowest-energy CO2 adsorption entries

surfaceComposition	facet	adsorption_energy	reactionEnergy	publication_title
CeO2-rocksalt-1Ovacancy	100	-2.843	-2.843	Selective high-temperature CO2 electrolysis enabled by oxidized carbon intermediates
SmCeO2-rocksalt-1Ovacancy	100	-2.751	-2.751	Selective high-temperature CO2 electrolysis enabled by oxidized carbon intermediates
CeO2-rocksalt-1Ovacancy	110	-2.067	-2.067	Selective high-temperature CO2 electrolysis enabled by oxidized carbon intermediates
CeO2-rocksalt-2Ovancies	111	-1.122	-1.122	Selective high-temperature CO2 electrolysis enabled by oxidized carbon intermediates
CeO2-rocksalt-1Ovacancy	111	-1.006 6	-1.006	Selective high-temperature CO2 electrolysis enabled by oxidized carbon intermediates

7 6. Competition between HCOO^* and COOH^* on metals

One of the most experimentally relevant questions is which branch of the first hydrogenation step is preferred. In this subset, four metals contain both HCOO^* and COOH^* entries. The interpretation

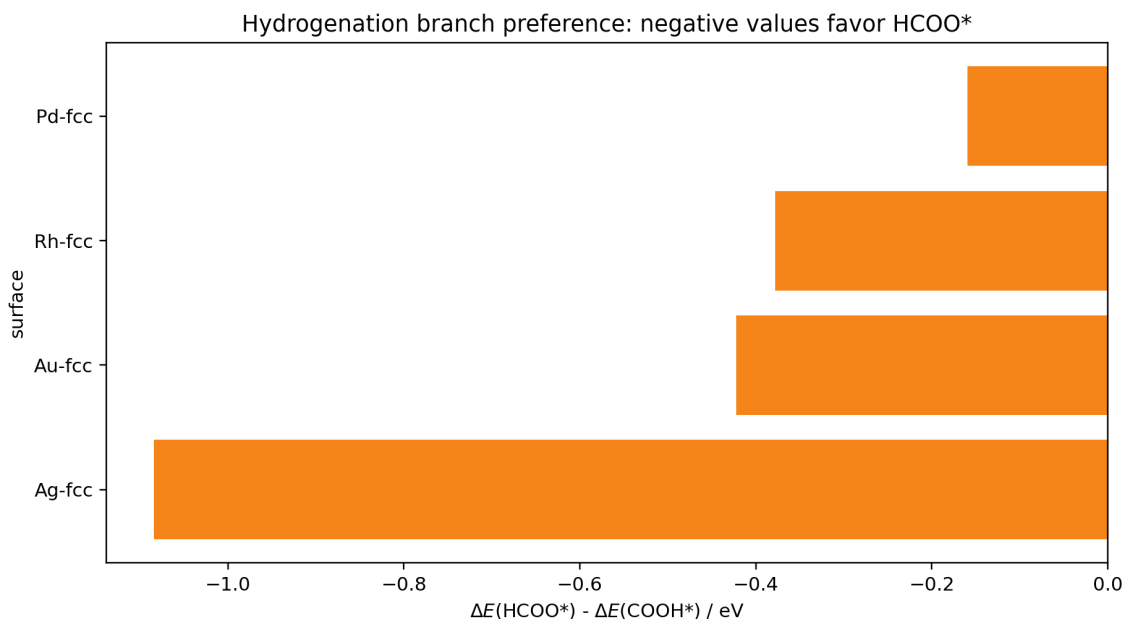


Figure 5: *
HCOO vs COOH

is straightforward:

- negative values mean HCOO^* is more stable than COOH^* ;
- positive values would mean the opposite.

Here, **all four metals favor HCOO^*** . The preference is strongest for Ag and weakest for Pd, but the sign is consistent across the available metal entries. For experiment, this suggests that under conditions represented by these calculations, the first hydrogenation event is more likely to populate a **formate-like intermediate** than a **carboxyl-like intermediate** on these metal surfaces.

7.1 Hydrogenation-branch comparison table

surfaceComposition	HCOO_minus_COOH_eV
Ag-fcc	-1.084
Au-fcc	-0.422
Rh-fcc	-0.378
Pd-fcc	-0.159

8 7. Direct CO_2 dissociation and available barriers

Barrier data are sparse in the downloaded subset, but they are highly useful because they immediately separate merely favorable final states from kinetically plausible steps. The barrier-containing subset leads to three practical takeaways:

1. **Ru(211)** stands out as the most favorable dissociation case in the downloaded data:

reaction energy -1.477 eV and a listed barrier very close to zero.

1. **Co(211)** is also favorable:

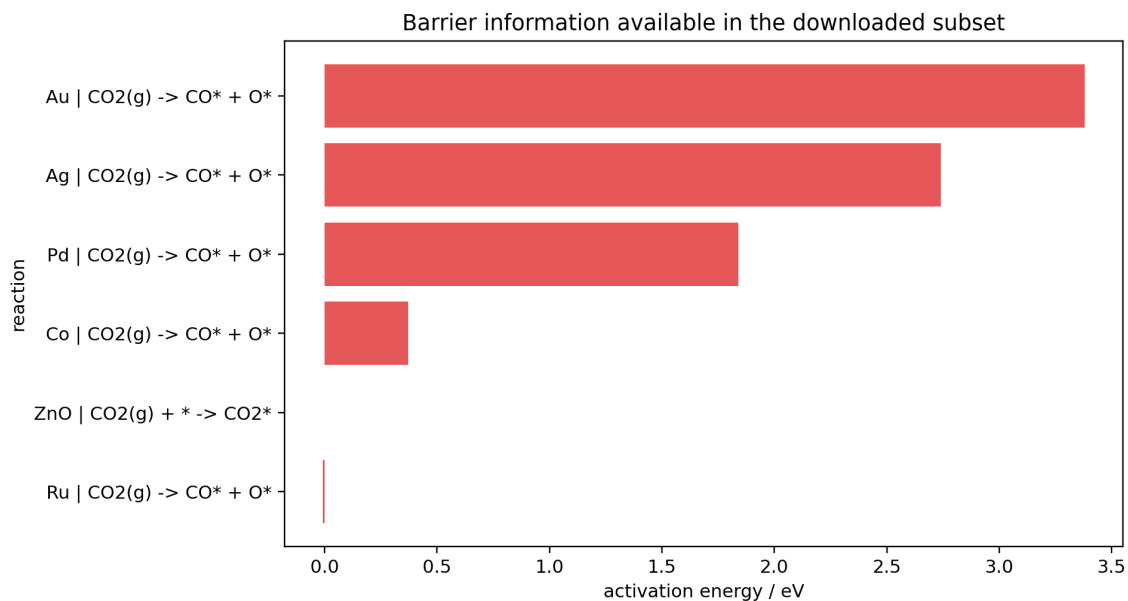


Figure 6: *
Barrier subset

reaction energy -1.107 eV with a moderate barrier of 0.372 eV.

1. **Pd(111)@@PLACEHOLDER0@@Ag(211)@@PLACEHOLDER1@@Au(111)‘ are much less promising for this direct dissociation step:**

both thermodynamics and barriers are less favorable. The ZnO entry is special because it corresponds to molecular CO₂ adsorption with a reported barrier of 0.000 eV, which is consistent with essentially barrierless adsorption in that specific study.

8.1 Barrier table

surfaceComposition	facet	Equation	reactionEnergy	activationEnergy	publication_title
Ru	211	$\text{CO}_2(\text{g}) \rightarrow \text{CO}^* + \text{O}^*$	-1.477	-0.009	Trends in CO oxidation rates for metal nanoparticles and close-packed, stepped, and kinked surfaces
ZnO	1	$\text{CO}_2(\text{g}) + * \rightarrow \text{CO}_2^*$	-1.04	0	Thermochemistry and microkinetic analysis of methanol synthesis on ZnO (0001)
Co	211	$\text{CO}_2(\text{g}) \rightarrow \text{CO}^* + \text{O}^*$	-1.107	0.372	Trends in CO oxidation rates for metal nanoparticles and close-packed, stepped, and kinked surfaces
Pd	111	$\text{CO}_2(\text{g}) \rightarrow \text{CO}^* + \text{O}^*$	0.74	1.84	Trends in the catalytic CO oxidation activity of nanoparticles
Ag	211	$\text{CO}_2(\text{g}) \rightarrow \text{CO}^* + \text{O}^*$	2.623	2.739	Trends in CO oxidation rates for metal nanoparticles and close-packed, stepped, and kinked surfaces
Au	111	$\text{CO}_2(\text{g}) \rightarrow \text{CO}^* + \text{O}^*$	3.29	3.38	Trends in the catalytic CO oxidation activity of nanoparticles

9 8. Most favorable and least favorable reactions in the downloaded subset

These two tables are a useful “bottom line” for rapid experimental reading. They show which elementary steps are strongly stabilized and which are strongly disfavored.

9.1 Ten most exergonic entries

surfaceComposition	facet	Equation	reactionEnergy	activationEnergy
CeO ₂ -rocksalt-1Ovacancy	100	$\text{CO}_2(\text{g}) + * \rightarrow \text{CO}_2^*$	-2.843	
SmCeO ₂ -rocksalt-1Ovacancy	100	$\text{CO}_2(\text{g}) + * \rightarrow \text{CO}_2^*$	-2.751	
CeO ₂ -rocksalt-1Ovacancy	110	$\text{CO}_2(\text{g}) + * \rightarrow \text{CO}_2^*$	-2.067	
Ru	211	$\text{CO}_2(\text{g}) \rightarrow \text{CO}^* + \text{O}^*$	-1.477	-0.009
CeO ₂ -rocksalt-2Ovacancies	111	$\text{CO}_2(\text{g}) + * \rightarrow \text{CO}_2^*$	-1.122	
Co	211	$\text{CO}_2(\text{g}) \rightarrow \text{CO}^* + \text{O}^*$	-1.107	0.372
ZnO	1	$\text{CO}_2(\text{g}) + * \rightarrow \text{CO}_2^*$	-1.04	0
CeO ₂ -rocksalt-1Ovacancy	111	$\text{CO}_2(\text{g}) + * \rightarrow \text{CO}_2^*$	-1.006	
CeO ₂ -rocksalt-1Ovacancy	111	$\text{CO}_2(\text{g}) + * \rightarrow \text{CO}_2^*$	-0.828	
Rh-fcc	211	$\text{CO}_2(\text{g}) + 0.5\text{H}_2(\text{g}) \rightarrow \text{HCOO}^*$	-0.823	

9.2 Ten most endergonic entries

surfaceComposition	facet	Equation	reactionEnergy	activationEnergy
CeO ₂ -rocksalt-1Ovacancy	111	$\text{CO}_2(\text{g}) - \text{O}_2(\text{g}) + * \rightarrow \text{C}^*$	5.778	
CeO ₂ -rocksalt	111	$\text{CO}_2(\text{g}) - \text{O}_2(\text{g}) + * \rightarrow \text{C}^*$	5.422	
CeO ₂ -rocksalt-2Ovacancies	111	$\text{CO}_2(\text{g}) - \text{O}_2(\text{g}) + * \rightarrow \text{C}^*$	5.395	
SmCeO ₂ -rocksalt-1Ovacancy	100	$\text{CO}_2(\text{g}) - \text{O}_2(\text{g}) + * \rightarrow \text{C}^*$	4.999	
CeO ₂ -rocksalt-1Ovacancy	100	$\text{CO}_2(\text{g}) - \text{O}_2(\text{g}) + * \rightarrow \text{C}^*$	4.961	
Ni-fcc	111	$\text{CO}_2(\text{g}) - \text{O}_2(\text{g}) + * \rightarrow \text{C}^*$	4.947	
CeO ₂ -rocksalt-1Ovacancy	110	$\text{CO}_2(\text{g}) - \text{O}_2(\text{g}) + * \rightarrow \text{C}^*$	4.859	
Ni-fcc	211	$\text{CO}_2(\text{g}) - \text{O}_2(\text{g}) + * \rightarrow \text{C}^*$	3.953	
Au	111	$\text{CO}_2(\text{g}) \rightarrow \text{CO}^* + \text{O}^*$	3.29	3.38
CeO ₂ -rocksalt	111	$\text{CO}_2(\text{g}) - 0.5\text{O}_2(\text{g}) + * \rightarrow \text{CO}^*$	3.055	

The most important chemical message from these extremes is that:

- **molecular CO₂ adsorption** can be strongly favorable on reduced ceria-type surfaces;
- ****C* formation**** is consistently too uphill to be considered an accessible first-step route in this subset.

10 9. What an experimental chemist can take away

If the question is “which classes of surfaces deserve attention for CO₂ activation?”, this dataset supports the following hierarchy:

- **Defective ceria-based oxides**: strongest CO₂ capture and activation in the present data.
- **Stepped transition-metal surfaces such as Ru(211) and Co(211)**: most promising for direct dissociation among the barrier-containing subset.
- **Au, Ag, Pd**: weaker for direct dissociation, but still chemically informative in hydrogenation-branch comparisons.
- **Ni**: intermediate behavior, with facet-sensitive CO₂ adsorption.

If the question is “which first hydrogenation branch is preferred?”, the answer from this subset is:

- the available metal entries consistently favor **HCOO*** over **COOH***.

If the question is “is direct carbon deposition from CO₂ a low-energy route here?”, the answer is:

- **no**. The C₀₂ → C* entries are far too uphill in this dataset.

11 10. Limitations of this dataset

This report should be read with a few clear limitations in mind:

1. The file is a **subset**, not the whole of Catalysis-Hub.
2. It mixes **different publications, codes, and functionals**.
3. The number of explicit **barrier entries is small**.
4. Most entries are **first-step thermodynamic snapshots**, not complete pathways.
5. Only a subset of rows can be used for exact adsorption-energy reconstruction because that requires the matching gas, clean surface, and adsorbate entries under the same reaction id.

These limitations do not erase the value of the data. They simply define the right use case: **comparative screening and mechanistic direction-finding**, not single-number absolute benchmarking across all systems.

12 11. Final conclusion

The downloaded Catalysis-Hub HDF already contains enough chemically resolved information to support a meaningful, master-thesis-level analysis for an experimental audience. The strongest and clearest results are:

- CO₂ adsorption is most favorable on defective ceria-derived oxide surfaces.
- Ni shows much weaker, facet-sensitive CO₂ adsorption.
- Ru(211) and Co(211) are the most favorable direct dissociation cases among the barrier-containing entries.

- The first hydrogenation step on the available close-packed metal surfaces favors HCOO^* over COOH^* .
- Direct formation of C^* from CO_2 is strongly disfavored in the downloaded subset.

In practical experimental language: this dataset points much more strongly toward **surface-specific CO_2 activation and selective first-step chemistry** than toward indiscriminate deep deoxygenation.